Exelon Generation

10CFR 50.73

CCN: 20-85

December 22, 2020

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Unit 2
Renewed Facility Operating License No. DPR-44
NRC Docket No. 50-277

Subject: Licensee Event Report (LER) 2-20-002

Enclosed is a Licensee Event Report which addresses a degraded condition due to Reactor Pressure Vessel (RPV) instrument nozzle leakage identified during a refueling outage at Peach Bottom Atomic Power Station Unit 2.

This LER is being submitted pursuant to the requirements of 50.73(a)(2)(ii)(A) for a Degraded Condition. There are no commitments contained in this letter. If you have any questions, please contact Matthew K. Rector at (717) 456-4351.

Respectfully,

Matthew J. Herr
Site Vice President
Peach Bottom Atomic Power Station

Enclosure

cc: US NRC, Administrator, Region I
US NRC, Senior Resident Inspector
W. DeHaas, Commonwealth of Pennsylvania
S. Seaman, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs

NRC FORM 366 (08-2020) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023
LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

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4. Title

Degraded Condition due to RPV Instrument Nozzle Leakage

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
10	29	2020	2020	- 002 -	00	12	22	2020		05000
									Facility Name	Docket Number 05000

9. Operating Mode	10. Power Level

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	[ ] 20.2203(a)(2)(vi)	[ ] 50.36(c)(2)	[ ] 50.73(a)(2)(iv)(A)	[ ] 50.73(a)(2)(x)
[ ] 20.2201(b)	[ ] 20.2203(a)(3)(i)	[ ] 50.46(a)(3)(ii)	[ ] 50.73(a)(2)(v)(A)	**10 CFR Part 73**
[ ] 20.2201(d)	[ ] 20.2203(a)(3)(ii)	[ ] 50.69(g)	[ ] 50.73(a)(2)(v)(B)	[ ] 73.71(a)(4)
[ ] 20.2203(a)(1)	[ ] 20.2203(a)(4)	[ ] 50.73(a)(2)(i)(A)	[ ] 50.73(a)(2)(v)(C)	[ ] 73.71(a)(5)
[ ] 20.2203(a)(2)(i)	**10 CFR Part 21**	[ ] 50.73(a)(2)(i)(B)	[ ] 50.73(a)(2)(v)(D)	[ ] 73.77(a)(1)(i)
[ ] 20.2203(a)(2)(ii)	[ ] 21.2(c)	[ ] 50.73(a)(2)(i)(C)	[ ] 50.73(a)(2)(vii)	[ ] 73.77(a)(2)(i)
[ ] 20.2203(a)(2)(iii)	**10 CFR Part 50**	[x] 50.73(a)(2)(ii)(A)	[ ] 50.73(a)(2)(viii)(A)	[ ] 73.77(a)(2)(ii)
[ ] 20.2203(a)(2)(iv)	[ ] 50.36(c)(1)(i)(A)	[ ] 50.73(a)(2)(ii)(B)	[ ] 50.73(a)(2)(viii)(B)	
[ ] 20.2203(a)(2)(v)	[ ] 50.36(c)(1)(ii)(A)	[ ] 50.73(a)(2)(iii)	[ ] 50.73(a)(2)(ix)(A)	

[ ] **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Matthew K. Rector, Manager - Regulatory Assurance	(717) 456-4351

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
[ ] No	[ ] Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

The event occurred on October 29, 2020, while Peach Bottom Unit 2 was performing Class 1 system leakage test following a shut down for refueling outage P2R23. Leakage was identified from a 2-inch Reactor Pressure Vessel (RPV) instrumentation nozzle (N-16A) during the test at a location where no leakage was found during the initial shutdown inspections. The leakage amount was identified as light seepage and visual examination revealed a surface crack at the 6 o'clock position on the interior RPV-instrument nozzle interface where the nozzle penetrates the vessel wall. Prior to de-pressurizing, an extent of condition bare metal examination was conducted on other instrument nozzles. No leakage was identified from any other instrument nozzles. Although the exact cause is unknown, phased array UT examination recorded a radial flaw in the N-16A J-Groove weld. A half nozzle repair was completed, consisting of replacing the outer portion of the existing nozzle with a new nozzle that was welded to the outside of the RPV. The reactor maintained shutdown until the pipe repairs and testing were completed. The safety significance of this event was minimal given the leakage was very small and was found while the reactor was shut down. Given the impact on the RPV boundary, this report is submitted in accordance with the requirements of 10 CFR 50.73 (a)(2)(ii)(A), which requires the reporting of any event or condition that resulted in the condition of the nuclear power plant, including its principal safety barriers, being degraded. On November 6, 2020, verbal approval for relief request I5R-14 regarding alternate repair of instrument penetration nozzle N-16A on the reactor vessel was received via email from J. Tobin (U.S. Nuclear Regulatory Commission) to D. Helker (Exelon Generating Company, LLC.) reference ADAMS: ML20314A028.

NARRATIVE

Unit Conditions Prior to the Event

Peach Bottom Atomic Power Station (PBAPS) Unit 2 was shut down for refueling outage P2R23 and in cold shutdown at the time of the event. The condition was identified during the RPV pressure test.

Description of Event

On October 29, 2020, at approximately 10:30 hours, while PBAPS Unit 2 was performing Class 1 system leakage test in cold shutdown, a through-wall leak was identified on instrument nozzle N-16A. The N-16A RPV penetration is a 2-inch instrument line nozzle located just above the Top-of-Active Fuel at 45-degree azimuth on the vessel.

The leakage was observed on the outer RPV wall during the performance of the system leakage testing being performed in accordance with ASME IWB-5210(a) requirements and station procedures. Visual examination revealed a surface crack at the 6 o'clock position and active leakage at the nozzle interface (annular gap) with the RPV outside diameter during the Class 1 system leakage test. Based on the design, any leakage into the annulus region between the nozzle body and the RPV wall would likely be associated with a through-wall failure of the nozzle assembly at the nozzle body.

Prior to de-pressurizing, an extent of condition bare metal examination (VT-2) of was performed on October 29, 2020, on the five other additional RPV instrument nozzles (N-11A, N-11B, N-12A, N-12B, and N-16B), and there was no evidence of leakage on any of the other instrument nozzles during the examination.

With the leakage confirmed to be originating from the N-16A nozzle assembly, a repair team was mobilized, consisting of station, corporate, and industry experts. The repair method involved installing a weld pad using Ambient Temperature Temper Bead (ATTB) welding in accordance with ASME Code Case N-638-7, "Similar and Dissimilar Metal Welding Using Ambient Temperature Machine GTAW Temper Bead Technique, Section XI, Division 1," and installing a half nozzle to the weld pad. This repair method will make the nozzle resistant to Intergranular Stress Corrosion Cracking (IGSCC).
The original partial penetration attachment weld and a remnant of the original nozzle will remain in place. The repair was completed on November 11, 2020, and the post-repair leakage test immediately followed, which validated the integrity of the repair and the RPV Class 1 pressure boundary.

On October 29, 2020, Emergency Notification System (ENS) # 54971 was made in accordance with 10 CFR 50.72 (b)(3)(ii)(A), for a degraded condition. Given the impact on the RPV boundary, this report is submitted in accordance with the requirements of 10 CFR 50.73 (a)(2)(ii)(A), which requires the reporting of any event or condition that resulted in the condition of the nuclear power plant, including its principal safety barriers, being seriously degraded.

Analysis of the Event

There was no actual safety consequence associated with this event. The potential safety consequence of this event was minimal, given the leakage was small and was found while the reactor was shut down for refueling. Plant Technical Specification (TS) 3.4.4 requires monitoring of reactor coolant leakage. When leakage limits are exceeded, the TSs requires reactor shutdown. If any leakage did exist during previous plant operation; it did not exceed TS leakage limits.

Prior to depressurizing from the RPV pressure test, an extent of condition bare metal examination (VT-2) on the five other

NARRATIVE

Analysis of the Event, continued

RPV instrument nozzles (N-11A, N-11B, N-12A, N-12B, and N-16B), and there was no evidence of leakage on any of the other instrument nozzles during the examination.

There are a total of six instrument nozzles including the N-16A nozzle, that are attached to the RPV with similar partial penetration (J-groove) welds on each unit. Each of the RPV penetrations are inspected for leakage each refueling outage during the RPV Class 1 system leakage test. Based on the most recent Class 1 system leakage test on October 29, 2020, Unit 2 Refueling Outage P2R23, there were no other RPV partial penetration welds exhibiting evidence of similar leakage.

A failure assessment and flaw evaluation were completed prior to startup to demonstrate the acceptability of leaving the original partial penetration attachment weld, with a maximum postulated flaw, in place for one-cycle. The flaw evaluation for the one-cycle was completed in accordance with IWB-3610 (LEFM method), as well as per Code Case N-749 (EPFM method).

Cause of the Event

Based on the results of the examinations performed of the N-16A nozzle (i.e., visual and ultrasonic), the most probable cause of the external leakage observed coming from the N-16A nozzle is that a single radial-axial oriented IGSCC flaw initiated in the J-groove weld and then propagated through the J-groove weld until it reached a depth where a leak path in the annulus between the nozzle and reactor vessel penetration existed.

A search of fabrication records for the N-16A nozzle and J-groove weld has not identified any anomalous material conditions or process deviations that could have contributed to the IGSCC indication observed; however, it is possible that subsurface fabrication defects could have existed to further propagate the flaw through the J-groove weld.

Corrective Actions

The following corrective actions have been completed:

A repair was completed to support one cycle of operation for Unit 2. The repair method involved installing a weld pad using ATTB welding in accordance with ASME Code Case N-638-7, "Similar and Dissimilar Metal Welding Using Ambient Temperature Machine GTAW Temper Bead Technique, Section XI, Division 1," and installing a half nozzle to the weld pad. This repair method will make the nozzle resistant to IGSCC. The original partial penetration attachment weld and a remnant of the original nozzle will remain in place. The repair was completed on November 11, 2020. The post-repair examinations performed in accordance with ASME Code requirements verified that there were no unacceptable indications in the newly installed weld pad or original base metal material.

A failure assessment and flaw evaluation were completed prior to startup to demonstrate the acceptability of leaving the original partial penetration attachment weld, with a maximum postulated flaw, in place for one operating cycle.

Relief Request (RR) 15R-14 was submitted to the NRC on November 4, 2020, due to the defect identified in the N-16A nozzle since a qualified technique to perform volumetric nondestructive examination (NDE) of the partial penetration weld for characterizing the flaw and determining flaw growth in the specific configuration does not exist, and to support repair efforts. This RR included information describing the cause of leakage, extent of condition, examination of the J-groove

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

Corrective Actions, continued

weld, flaw analytical evaluation, repair of N-16A nozzle, corrosion evaluation, and loose parts evaluation. On November 6, 2020, verbal approval was granted via email from J. Tobin (U.S. Nuclear Regulatory Commission) to D. Helker (Exelon Generating Company, LLC), "Peach Bottom Verbal Relief for Penetration Nozzle (EPID: L-2020-LLR-0144)," dated November 6, 2020 (ML203114A028). The Relief Request 15R-14 is for Unit 2 operating cycle 24, which is scheduled to end in the Fall of 2022.

The following corrective actions are being planned:

A separate RR will be submitted to justify continued use of the nozzle repair for the life of the plant. This permanent RR, which will contain the appropriate analyses and justification for the remainder of the plant operating life, will be submitted prior to the end of the current operating cycle (Cycle 24).

Previous Similar Occurrences: 2012 - Quad Cities Unit 2 (N11B); 2017 - Limerick Unit 2 (N16D)